

COMPUTER NETWORKS — WEEK 2 CHEAT SHEET

PPP · ATM · IEEE 802 Standards · CSMA/CD · Token Bus · Token Ring | Instructor: D. MSI

1. Data Link Layer in the Internet

The Internet is made of **hosts** (laptops) and **routers** (forward data). They connect via LAN or point-to-point lines.

Point-to-point lines used when:

- One LAN connects to another LAN (like a leased line)
- A home user dials up to the Internet

Both cases use **PPP** (Point-to-Point Protocol) as the data link protocol.

2. PPP — Point-to-Point Protocol

PPP connects two devices directly (e.g., home router to ISP router). Based on HDLC frame format. Runs in **unnumbered mode** (no seq numbers, no ACKs).

3 Key Features

- **Framing:** defines frame structure + error check
- **LCP:** sets up, tests, and closes the link
- **NCP:** agrees network layer settings (one NCP per protocol: IP, IPX, AppleTalk)

PPP Frame Fields

Field	Value	Meaning
Flag	01111110	Start/end marker
Address	11111111	Broadcast, always same
Control	00000011	Unnumbered frame
Protocol	2 bytes	0=network data, 1=control
Payload	≤1500 B	Actual data
Checksum	2 B (default)	Error check

PPP Phases (in order)

Dead → Establish (LCP) → Authenticate (optional) → Network (NCP) → Open (data flows) → Terminate

Trick: Dead-Establish-Auth-Network-Open-Terminate = a link's full life story

3. LCP Frame Types

- **Configure-Request/Ack/Nak/Reject** — set up link options
- **Terminate-Request/Ack** — close the link
- **Code-Reject / Protocol-Reject** — unknown code or protocol
- **Echo-Request / Echo-Reply** — like "ping", checks link is alive
- **Discard-Request** — for testing only

Nak = "no, try this instead". Reject = "no, completely refused".

4. ATM — Asynchronous Transfer Mode

ATM sends data in small fixed-size packets called **cells** = 53 bytes (5-byte header + 48-byte payload). The 48-byte size was a political compromise, not technical.

Header Fields

Field	Size	Job
GFC	4 bits (UNI only)	Local flow control
VPI	8b(UNI)/12b(NNI)	Virtual path route
VCI	16 bits	Virtual channel route
PT	3 bits	Data or OAM cell
CLP	1 bit	1=can be dropped
HEC	8 bits	Checks header errors

UNI = user device to ATM switch (has GFC). **NNI** = switch to switch (no GFC, bigger VPI).

Trick: ATM cell = 53 = 5 (head) + 48 (data). CLP=1 → can drop.

5. Why IEEE 802 Standards?

Before standards: no compatibility, no mass production, high prices, low adoption. IEEE formed a committee to fix this using numbering **802.x**.

Standard	Covers
802.1	Internetworking
802.2	Logical Link Control
802.3	Ethernet (CSMA/CD)
802.4	Token Bus LAN
802.5	Token Ring LAN
802.6	MAN
802.11	Wireless (Wi-Fi)
802.16	Broadband Wireless (WiMAX)

802.11 Wi-Fi Versions

Standard	Speed	Freq
802.11	1-2 Mbps	2.4 GHz
802.11a	54 Mbps	5 GHz
802.11b	11 Mbps	2.4 GHz
802.11g	20-108 Mbps	2.4 GHz
802.11n	Highest	2.4 & 5 GHz

6. IEEE 802.3 — CSMA/CD (Ethernet)

Used in bus LANs to share one cable. Has 2 parts:

CSMA (listen first)

- Station checks if bus is free ("carrier sense")
- If free → send now. If busy → wait random time, try again

CD (handle collision)

- Two stations send at the same time → collision
- CD detects it, both stop immediately
- Both wait random time, then retry

Standard	Speed
802.3	10 Mbps
802.3u	100 Mbps (Fast Ethernet)
802.3z	1 Gbps (Gigabit)
802.3ab	10 Gbps

Trick: CSMA = Listen before you speak. CD = Stop if collision.

7. IEEE 802.4 — Token Bus

A **token** (control packet) passes station to station on a bus. Only the token holder may send data — this stops collisions completely.

Steps

- Token moves node to node on the bus
- Station with data grabs empty token, attaches data
- Token travels to destination, delivers data
- Receiver sends ACK back; token passes on
- Error → retransmission request attached to token

Physical layout = bus, but token order = logical ring.

8. IEEE 802.5 — Token Ring

Devices sit in a **ring**. A token circulates non-stop; only the holder can transmit — no collisions.

Steps

- Station waits for a free (empty) token
- Attaches: sender address, receiver address, data, CRC
- Token travels the ring to the destination
- Receiver checks errors, sends ACK or asks resend

Used in: IBM Token-Ring LANs, and FDDI (fiber-optic ring backbones).

Trick: Token Bus=802.4, Token Ring=802.5, FDDI=fiber ring

Quick Comparison & Exam Recall

Protocol	IEEE	Topology	Collision?
Ethernet	802.3	Bus	Yes (detected)
Token Bus	802.4	Bus	No
Token Ring	802.5	Ring	No
Wi-Fi	802.11	Wireless	Avoided
ATM	—	Switched	No (cells)

- PPP flag=01111110, address=11111111, control=00000011
- ATM cell=53B (5 header + 48 payload); UNI has GFC, NNI doesn't
- CSMA/CD = listen then send, stop & retry on collision
- Token Bus & Token Ring both avoid collisions using a token

বল স্কলন: PPP দুটি ডিভাইসকে সরাসরি সংযুক্ত করে, HDLC ফরম্যাট মেনে চলে। ATM ভেটা পাঠায় ৫৩ বাইটের ছোট cell এ। IEEE 802 স্ট্যান্ডার্ড ডিভাইসগুলোর মধ্যে সামঞ্জস্য আনে। CSMA/CD ইথারনেটে সংঘর্ষ সামলায়। Token Bus ও Token Ring টোকেন ব্যবহার করে সংঘর্ষ এড়ায়।